L1.5: Determinants (part 2)

Transcript Language

English

Hello and welcome to the Maths 2 component of the online BSC program

on data science. Today's video is on determinants, this is the second part of

the videos on determinants. So, let us recall first what we did in part 1.

So, recall from part 1, that we defined the determinant of a 1 by 1 matrix so if a is singleton a then determinant of A is just that number a.

If A is the 2 by 2 matrix a, b, c, d, then determinant of A is a d minus b c you multiply

the diagonal terms and subtract out the off diagonal terms and if you have a 3 by 3 matrix,

then we defined the determinant by what we call expanding with respect to the first row.

So, what that meant was you take the first the 1 1 entry of the first row, the first entry of the

first row and then disregard everything in the first row and the first column

you get a 2 by 2 matrix remaining take its determinant and multiply that with a 1 1

that is the first term. For the second term you look at a 1 2

and do the same process disregard everything in the first row and the second column,

multiply the determinant of the resulting 2 by 2 matrix with a 1 2 and for the third term

you look at a 1 3 disregard everything in the first row and the third column,

take the determinant of the resulting 2 by 2 matrix and multiply it with a 1 3.

And then you alternately add these meaning take the first expression minus the second expression

plus the third expression. So, that was the determinant of a. So,

the determinant of the 3 by 3 matrix was defined in terms of determinants for 2 by 2 matrices

and then we can expand this out we can work out explicitly what the expression is and you get some

rather long and nasty 6 term expression.

So, let us do an example just to refresh ourselves maybe a 3 by 3 example.

So, here is a 3 by 3 matrix 2 4 3, 0 8 7, 0 0 9 so maybe I should mention that such

a matrix is called an upper triangular matrix because it has zeros below the diagonal.

So, let us compute its determinant, so determinant of A is 2 times the matrix 8 7 0 9 which is what

is remaining if you disregard everything in the first row and column minus 4 times determinant

of 0 7 0 9, plus 3 times determinant of 0 8 0 0 and then if we work out what these are

using the determinants of 2 by 2 matrices, we will get 2 times

8 times 9 minus 7 times 0 which is 72 minus 0 minus 4 times 0 minus 0 plus 3 times 0 minus 0,

so this is just 2 times 72 minus 4 times 0 plus 3 times 0 which is just 2 times 72 which is 144

and I will ask you to notice something important which is that this is 2 times 8 times 9

which is exactly what we have on the diagonal. So, these are the diagonal entries over here and

that is what we got, so the determinant of this matrix A is the product of the diagonal entries.

So, I hope you followed both the computation and the fact that it is

the product of the diagonal entries. So, this is not some

fact specific to this example, this is a general fact for what we call any upper

triangular matrix or any lower triangular matrix. So, for such matrices the determinant is the

product of the diagonal elements. So, I will just quickly recall what

is an upper triangular matrix as I said in the beginning. So, an upper triangular matrix means

if you so this is a general matrix of size n by n, so this is called upper triangular if you look at the diagonal and everything below the diagonal is 0. So, all these entries are zeros.

So, we are going to write this as a i j is 0 if i is greater than j. So, if the matrix satisfies

this property it is an upper triangular matrix and this is how it looks like.

Similarly, so then so this is upper triangular. So, similarly we have lower triangular, so a lower

triangular matrix. So, this is again n by n, that is if a i j is 0

if now this time i is strictly less than j. So, how does that look like, not a table but a matrix,

so this part must be 0 and these elements can be anything that any numbers. So, for such matrices

the determinant is the product of the diagonal elements. So, of course we know this only for

3 by 3 and I will suggest that you check this for 3 by 3 and 2 by 2 matrices.

So, let us move on to a special kind of matrix namely if you have a matrix then its transpose,

so let us define the transpose and compute its determinant.

So, in general if you have an m by n matrix, the transpose of this matrix is a new matrix of size n by m so it has n rows and m columns. So, remember the first one had m rows and n

columns a and m rows in columns, so a transpose has n rows and m columns and its ij-th entry

is a j i. So, the ij-th entry of a transpose is whatever was the jith entry of A. So just,

this is the notation it is called A superscript T, so T is for transpose and its definition

as we noted is A transpose i j is A j i. So, just for example if A is a 3 by 3 matrix

then a transpose is again 3 by 3. So, square matrices the transpose has the

same size it is also a square matrix of the same size and what happens to the entries,

so if A is like this a 1 1 a 1 2 a 1 3 and so on then a transpose is a 1 1, a 2 1, a 3 1, a 1 2,

a 2 2, a 3 2, a 1 3, a 2 3, a 3 3. So, what happened here these two entries got interchanged,

these two entries got interchanged, so the diagonal remains the same.

So, it reflects about the diagonal if you prefer a more geometric

way of thinking about it. So, that is what happens in the transpose. So, what do we want to do now?

We want to compute what is the determinant of the transpose. So, let us work this out. So,

determinant of a transpose, so we have a clear formula for the determinant so let us use that.

So, we look at the 1 1 term of A transpose and we multiply by the determinant of what

is left when we drop the first row and column and minus the 1 2 term of the transpose times

determinant of whatever is left when we drop the first row and second column of A transpose

and plus the 1 3 term of A transpose times determinant of whatever is left when you drop the

first row and the third column of A transpose. So, if we do that then this is exactly the

expression we get and now we can work out explicitly what happens

and you will see that the final expression you get is just some is just the same as the expression

for determinant of A except that the terms have changed places but it is the same expression.

So, this is just determinant of A. So, what is the upshot? If you take the transpose of A square matrix, then its determinant remains unchanged and we have in fact proved this in the 3 by 3 case

I will request that you do it for the 2 by 2 as well, 1 by 1 there is of course nothing to do.

So, now let us move on to minors and cofactors. So, this is a very important

idea and this is what is going to help us to generalize the determinant to the n by n

situation meaning where the size of the square matrix is n by n.

So, now suppose A is an n by n square matrix where n is at most 4 we will do it in this

particular case so n is so either it is a 2 by 2 matrix or a 3 by 3 matrix or a 4 by 4 matrix.

Then the minor of the entry in the ith row and jth column is the determinant of the sub

matrix formed by deleting the ith row and the jth column. So, we have been doing this

process in computing the determinant, so to compute the determinant of the 3 by 3 matrix

for the first expression in the determinant we take a 1 1 times determinant of the 2 by

2 matrix which is obtained by deleting the first row and first column. So, now we are talking about

just that part, just that operation. So, you look at the ith row and the jth

column and you delete those and then you get a matrix of size n minus 1 by n minus 1

and then you look at its determinant. So, now this makes sense because we have defined

determinants for 2 by 2, 1 by 1, 2 by 2 and 3 by 3 matrices. So, if you take a matrix of size 4 by 4

and you delete the ith row and the jth column you will get matrix of size 3 by 3 and then

you take its determinant. So, this is called the ij-th minor and its denoted by m i j.

So, here is an example, I said a lot of words so here is a explicit example.

So if a is a 3 by 3 matrix as written here a 1 1, a 1 2, a 1 3, a 2 1, a 2 2, a 2 3 and a 3 1,

a 3 2, a 3 ,3 then the 1 1 minor the 1 1th minor is nothing but determinant of a 2 2, a 2 3, a 3 2,

a 3 3. So, this is what you obtain by dropping the first row and first column, this is exactly the

kind of operation we have been using in computing determinants by expansion along the first row and

let us go ahead and first define what is the ij-th cofactor.

So, the ij-th cofactor is where you take the ij-th minor and multiply it by a minus 1 to

the power i plus j. Why is the sign coming, so as we saw in the expression for the determinant

there was an alternating sign, the first term was with a plus then you subtracted the second

term then you added the third term, so it was alternating. So, somehow this minus 1 to the i

plus j is supposed to take care of that sign that is why we have the sign over here. So,

it is the same number but it could be with a minus sign depending on what is i and what is j.

So, for example C 1 1 is just minus 1 to the power 1 plus 1 which is minus 1 to the power 2, so

minus 1 squared is 1 times M 1 1, so it is minus 1 square times M 1 1 which is just M 1 1. So,

as a similar in similar light if we have C 2 3, then this is minus 1 to the 2 plus 3 times M 2 3,

but now 2 plus 3 is 5 which is odd so minus 1 to the power 5 is minus 1. So, this is minus M 2 3.

So, it depends on what is i and j accordingly we will pick up a minus sign.

So, I hope it is clear what the cofactors and minors are.

So, let us now try to rewrite the determinant in terms of minus and cofactors.

So, for a 3 by 3 matrix so we have determinant of AA is this is expansion along the first row,

this is the definition a 1 1 times determinant of the matrix obtained by deleting the first row

and column and then you take its determinant but that is exactly what we called the minor

and then minus a 1 2 times determinant of a 2 1, a 2 3, a 3 1, a 3 3, so you deleted the first row

and second column and then took the determinant but that is exactly what we call the 1 2th minor

and then similarly for the third term. So, we can rewrite it now in terms of our

minus as a 1 1 times M 1 1 minus a 1 2 times M 1 2 plus a 1 3 times M 1 3. So, we are just

rewriting these determinants as minus this was the definition if you remember from the previous slide

and then if I want to take care of this sign, there is a minus sign here and I want to write

everything as a plus so then I can rewrite this as a 1 1 times the 1 1th cofactor plus a 1 2 times the 1 2th cofactor plus a 1 3 times the 1 3th cofactor. So, that is why we introduce cofactors

if we want to not remember the signs, then we will take care of those signs using the cofactors.

So, this is the expression for the determinant of a 3 by 3 matrix in terms of the minors

and remember the minors meant we or the cofactors the minors meant that you compute determinants of

2 by 2 matrices. So, now we can go beyond, so maybe first we should remark that this formula

works even for a 2 by 2 matrix, so if you take a 2 by 2 matrix say a b c d, then the determinant is

a d minus b c so we can rewrite that as A times M 1 1 minus B times M 1 2, so C is exactly

M 1 2 and d is exactly M 1 1 or if you want a plus sign then we can write it as a times C 1

1 plus b times C 1 2, so check this. So, now we can use this to generalize

beyond n is 3. So, let us generalize this to A 4 by 4 so suppose you have a 4 by 4 matrix now

then how do we define the determinant. So, the determinant is this is expansion along the first row so the determinant is you take minus 1 to the power 1 plus j because we have the first row here

a 1 j times M 1 j, this is exactly the expression that we have up here this is exactly this

expression here, except that this is for 3 by 3 I am writing the same expression for 4 by 4.

And if you want to take care of the signs not bother about the signs then you can write it just

as a sum of a 1 j times the corresponding cofactor for the 1 jth term. So, let us

maybe write this expression down explicitly. So, what does this mean?

This means that the determinant of so for a 4 by 4 matrix so a 1 1 times determinant of

whatever you get after dropping the first row and the first column so a 2 2, a 2 2, a 2 3, a 2 4,

a 3 2, a 3 3, a 3 4 and then a

4 2, a 4 3, a 4 4, that is your first expression, the first term in this sum and then minus

a 1 2 and I will maybe write one more term there should be a determinant here determinant of so now

we drop the first row and the second column. So, now we get a 2 1, a 2 3, a 2 4, a 3 1, a 3 3, a 3 4 and then a 4 1, a 4 3, a 4 4. That is the second term in this expression

and note we have a minus sign here and then plus a 1 3 and I hope now you can write down

these two determinants. So, check out what this is and then minus a 1 4 times determinant of

whatever we get here. So, check these two terms see if you can write down these two terms

that will mean you have understood what is going on.

So, what is the point here? The point is we could write the determinant of a we defined it

this is a 4 by 4 matrix in terms of determinants for 3 by 3 matrices. So,

in other words if I know how to define the determinant for a matrix of smaller size then

I can sort of use this process inductively and generalize and that is exactly what we do.

So, here is the inductive definition of the determinant. So, suppose you have a matrix of size

n by n and we know how to define determinants for size n minus 1, so n minus 1 by n minus 1 matrices

we know how to define the determinant. So, then we will define miners and cofactors as before, so what does that mean? The ij-th minor is the determinant of the sub matrix formed by deleting

the ith row in the jth column. So, we delete the ith row and the jth column we get a matrix

of size n minus 1 by n minus 1 and we look at its determinant that is called the ij-th minor.

This makes sense because we have assumed that we know how to define matrices, determinants for n

minus 1 by n minus 1 matrices and then if you want to again we want to do something similar to what

we did earlier, so if you do not want to bother about signs going ahead you define the ij-th

cofactor as minus 1 to the power i plus j times the ij-th minor and then we define the determinant

of A as the again this is by expansion along the first row as minus 1 to the power 1 plus j.

Now the sum runs from 1 through n because remember there are n entries in the first row,

so minus 1 to the power 1 plus j a 1 j times m 1 j and again if you do not want to bother about signs

and you just want a simple sum then you absorb that sign into the minor and you can write it in terms of cofactor, so summation a 1 j, C 1 j. So, maybe let us write down this expression let me

write down the first term, so now suppose let us say I have a n by n matrix so this determinant of

A is a 1 1 times determinant of whatever you get by deleting the first row and the first column.

So, you will get a 2 2, a 2 3 and it will run all the way till a to n and then below you will have

a 3 2, a 3 3 and all the way up to a three n and if you complete this you get up go all the

way up to a n n and then the second term is going to be minus a 1 2 times determinant of

a 1 not a 1, a 2 1, a 2 3. And then you keep going up till a to

n and then you have a 3 1, a 3 3 going up till a 3 n and you keep doing this here you will get the

first column, here you will get the third column meaning all the entries remaining in third column

and here you will get the nth column and then you keep going, you have a 1 2, then you have a 1

3 times determinant see if you can write down this term and then minus a 1 4 times some determinant

and note what is happening. These determinants are of smaller size.

So, now if I want to do this for a 5 by 5 matrix, I will need to compute of the determinant of a 4

by 4 matrix but I know how to do that, which means I now know how to compute the determinant of a 5

by 5 matrix but then I can use the same inductive definition and go for a 6 by 6 matrix and then

using 6 by 6 to 7 by 7 and then so on to 8 by 8 and then using that to 9 by 9 and you can see that

inductively for any number I can do this process. So, that is the idea of the determinant. So,

this is a very complicated looking definition and it is kind of a quirk that it is indeed very

useful I in fact I cannot emphasize how useful you will see it later when we solve equations

and it is used for other things as well you will see this in calculus too.

So, maybe as an example let us compute the determinant of the identity matrix of size n.

So, remember the identity matrix had 1 on the diagonal so and everything else is 0.

So, I am this is shorthand to say that everything else is 0.

So, let us use the definition, so by definition this is 1 times determinant of whatever the 1 1th minor, 1 times the 1 1th minor. So determinant of you drop the first row

and first column so you get ones on the diagonal and zeros on the off diagonal.

So, this is again the identity matrix of size n minus 1 by n minus 1

and then minus 0 times the 1 2th minor but I do not want I do not really care about

computing it because it is 0 times something it is going to be 0 anyway plus 0 times the

1 3th minor and so on. So, all these terms are going to be 0 because in the rest of the first

row all the terms you have are 0, so what do you get, you are left only with 1 times determinant

of the identity matrix of smaller size. So, determinant of In is determinant of

In minus 1 that is what we obtained in by this process and then I can keep doing this. So,

this is determinant of In minus 2 and so on. So, the correct way of doing this is what is called

induction and doing this we will reach all the way down till maybe I3 which in or I2

whichever you prefer and both of these we computed in the previous video to be 1.

So, let us recall what we have done in this video we started by defining recalling the definition of

the determinant for the 1 by 1, 2 by 2, and 3 by 3 cases and we saw that for the 4 by 4

matrix we can define the determinant by using the 3 by 3 definition because we can do that

for the 3 by 3 using the 2 by 2 and for the 2 by 2 using the 1 by 1. So, it follows that for

the 4 by 4 we can use the 3 by 3 definition and then we generalize this to the n by n matrix,

so we said that if we can define the determinant for n minus 1 by n minus 1 matrix, then we can

define it for an n by n matrix so, this is called the inductive definition of the determinant.

And finally, we have computed we did one example which was we computed that the determinant of

the identity matrix of any size 3 by 3, 4 by 4, 5 by 5, 10 by 10; any n by n is 1, thank you.